

TEMPORARY HALF-PIPE RAMP - WIND AND HOLD-DOWN

Ref TW-HP-001 Rev P01 4 October 2026 | Banksy, for Dal Hopwood

NOT A DESIGN - NOT FOR CONSTRUCTION - NOT A PERMIT TO LOAD

1. WHAT THIS IS

NOT A DESIGN - NOT FOR CONSTRUCTION - NOT A PERMIT TO LOAD OR RIDE.

The wind pressure is calculated and every step is shown. The STRUCTURE is not designed here.

The design and its independent check must be produced, checked and OWNED by a competent

temporary works designer, with the structural verification to the governing standards below.

Nothing is erected, ballasted or ridden until a permit to load has been issued.

It calculates the wind pressure at the site and what it does to a 20 m ramp, and it sets out the hold-down demand. It does NOT design the structure, the transitions or the connections, and it does not cover the loads a skateboarder puts into it.

2. THE STRUCTURE AND THE BASIS

Item	Temporary half-pipe, 20.00 m long, 5.00 m riding surface, coping 2.00 m high
Cross section	1.00 m flat bottom + two 2.00 m transitions, decks 0.60 m each side
Footprint	20.00 m x 6.20 m overall
Exposure	EXTERNAL and open to the sky - a full wind case applies
Setting	town assumed; the site is not confirmed
Duration	cprob = 1.0 (more than one year)
Wind area	40.0 m2 across the length - the ramp is a sail

3. WIND PRESSURE - AND WHY THIS IS THE POINT OF THE WHOLE PACK

Dynamic pressure $q_p = 0.613 \times c_{prob}^2 \times C_{ef} \times S_{wind}^2$, with S_{wind} and C_{ef} taken from the validated TWf2012:01 calculator that reproduces the published worked example. Those two are properties of the SITE and the HEIGHT only - they do not depend on what is being checked, so they carry across to a ramp. The hoarding's shape coefficients do NOT, and are not used here.

vb,map	from sea	Cef	Swind	qp kN/m2
22 m/s	2 km	1.070	22.44	0.330
22 m/s	30 km	0.994	22.44	0.307
25 m/s	2 km	1.070	25.50	0.427
25 m/s	30 km	0.994	25.50	0.396

Governing for what follows: $q_p = 0.427 \text{ kN/m}^2$, the worst row. The range is shown because the site is not yet confirmed.

4. WHAT THE WIND DOES TO IT

The ramp presents its 40 m² side elevation to a cross wind - 20.00 m long by 2.00 m high. The horizontal force is $q_p \times C_f \times A$. C_f is NOT from any standard I hold: it is assumed, and carried as a sensitivity so the designer can see what their own figure costs.

Cf assumed	wind force kN	sliding resist. at mu 0.30 kN	extra ballast t	verdict
1.0	18.7	62.4	3.0	ballast
1.3	24.3	81.1	4.9	ballast
1.6	30.0	99.8	6.8	ballast

Self weight is about 3.4 t (ply, timber, rib plates), which at $\mu = 0.30$ resists about 10 kN on its own - so the ramp needs holding down on top of its own weight in every case. That is the commercial headline: ballast in the order of 3 to 7 tonnes, or ground anchors, before anyone rides it.

Uplift is not calculated here. A trough is open on top: wind over the opening can lift the surface, and wind getting under an unclosed side can do worse. Both are the designer's checks - the closed sides shown on the sheet are there precisely so wind cannot get inside.

5. THE STANDARDS THAT GOVERN IT

BS EN 14974:2019 Skateparks - safety requirements and test methods. THE product standard for a facility used by the public on skateboards, roller sports and BMX. Supersedes BS EN 14974:2006+A1:2010.

BS EN 13814-1:2019+A1:2024 Safety of amusement rides and amusement devices - design and manufacture. Applies to mobile and temporary devices: if the ramp is hired in as an attraction this is the design standard.

BS EN 1991-1-4 Wind actions, where the designer works to Eurocode rather than the simplified method.

BS 5975-1:2024 / BS 5975-2:2024 Temporary works procedure: roles, register, check categories, permits. Replaced BS 5975:2019.

CDM 2015 Principal Contractor duties, the design brief, and the temporary works register.

BS EN 1090-1 Execution of steel structures - if the coping, anchors or frame are fabricated steel.

6. NOT APPLICABLE, SAID OUT LOUD

- The Temporary Works Forum hoarding guide (TWf2012:01) does NOT govern this. Only its wind pressure route is borrowed, and said so above.

- Building Regulations do not govern a temporary ramp.

- Highways Act 1980 s.169 does not apply if the ramp is inside a car park or yard.

7. OPEN INPUTS - NOTHING IS BOUGHT OR ERECTED UNTIL THESE CLOSE

1. THE SITE. The wind map value comes from BS 5975-2 Figure 8 for the actual location; the

range above is shown because the site is not confirmed. Also: town or country, altitude,

and distance to the sea.

2. THE SHAPE FACTOR. Not taken from any standard I have. It is assumed and carried as a

sensitivity - 1.0 to 1.6 covers the range. The designer sets it, and it drives the ballast.

3. THE HEIGHT. Adopted 2.00 m to the coping, because $5.00\text{ m} = 1.00\text{ flat} + 2 \times 2.00\text{ transition}$.

If Dal wants a different height the whole pack re-runs - it is one number.

4. THE GROUND. Tarmac assumed. What is under it decides whether ground anchors are viable

instead of ballast, and it changes the price.

5. PUBLIC USE. Whether this is ridden by the public, supervised, or event-only, decides which

of the two product standards applies and how heavy the inspection regime is.

6. DYNAMIC LOADS. A skateboarder landing on a transition is a dynamic action. Not modelled

here at all - the designer's to derive.

A wind check is not a permit to load. A ramp is not ridden until the design, the hold-down and the ride inspection have been signed off.